

Tracing the Unseen: Trajectory Recovery from Cellular Records

(Online Appendix)

Jingyu Zhu [†], Yu Sun [†], Shaoxu Song [§], Dewang Ren [‡], Junhui Liu [‡], Xiaojie Yuan [†]

[†] Nankai University [§] Tsinghua University [‡] Xi'an Aeronautics Computing Technique Research Institute
 {21202407094@mail., sunyu@, yuanxj@}nankai.edu.cn, sxsong@tsinghua.edu.cn, {rendw007, liujh200}@avic.com

This document is the online appendix of the paper “*Tracing the Unseen: Trajectory Recovery from Cellular Records*”. It provides the complete proofs of all theoretical results. Propositions and lemmas are numbered separately, following the main paper, and each result is restated below before its proof.

Notation. $\mathcal{M}$ denotes the road network and $d_{\mathcal{M}}(\cdot, \cdot)$ the shortest-path (network) distance. $\mathcal{P}$ is the set of candidate locations, $\mathcal{R} \subseteq \mathcal{P}$ the set of representative candidates with $r = |\mathcal{R}|$, and n the number of timestamps $t_1, \dots, t_n$ of a cellular record sequence. $\Omega_i(p)$ is the coverage count of location p at timestamp t_i , i.e., the number of observed base stations whose coverage regions contain p ; the main paper defines $\Omega_i(p)$ as the *set* of such stations and this appendix uses its cardinality throughout. $\Omega(P) = \sum_{i=1}^n \Omega_i(p_i)$ is the overall coverage of a trajectory $P = \{p_1, \dots, p_n\}$, $C_i^{\max} = \max_{p \in \mathcal{P}} \Omega_i(p)$, and $\Omega_{\max} = \max_{i \in [1, n], p \in \mathcal{P}} \Omega_i(p)$, so that $\Omega(P^*) \geq \Omega_{\max}$. D_c is the coverage region of base station c (the set of candidates it covers), $U = \bigcup_{i=1}^n \bigcup_{c \in C_i} D_c$ the union of all observed coverage regions, R the minimum axis-aligned bounding rectangle enclosing U , and ∂R the boundary of R ; $\mathcal{M}'$ is the localized subnetwork built from R and ∂R as in the main paper. A transition (p_i, p_{i+1}) is *reachable* if $d_{\mathcal{M}}(p_i, p_{i+1}) \leq s \cdot (t_{i+1} - t_i)$. s is the speed constraint, δ the group diameter, and g the candidate granularity (the fixed sampling resolution of $\mathcal{P}$ along road edges). TREC, TRRP, TRHA, and TRSN refer to Algorithms 1–4 of the main paper, respectively. The symbol θ indexes candidate groups in TRRP (Lemma 1 and Propositions 3–4) and subnetworks in TRSN (Proposition 8); the two uses are contextually separate and never occur in the same statement. This appendix is a standalone document that compiles independently of the main paper, following the main paper’s numbering.

TREC is the exact dynamic programming algorithm based on the recurrence

$$\omega_{ij} = \begin{cases} \Omega_1(p_j) & i = 1 \\ \max_{\substack{p_k \in \mathcal{P} \\ d_{\mathcal{M}}(p_j, p_k) \leq s \cdot (t_i - t_{i-1})}} \{\omega_{i-1, k} + \Omega_i(p_j)\} & i > 1, \end{cases} \quad (1)$$

where ω_{ij} is the maximum cumulative coverage achievable from t_1 to t_i when the device is located at candidate p_j . TRHA

greedily selects, at each timestamp t_i , a feasible candidate maximizing the score

$$S_i(p) = \alpha \cdot \frac{\Omega_i(p)}{C_i^{\max}} + (1 - \alpha) \cdot \left(1 - \min \left\{ 1, \frac{d_{\mathcal{M}}(p, \bar{p}_{i+k})}{s \cdot (t_{i+k} - t_i)} \right\} \right), \quad (2)$$

where $\bar{p}_{i+k}$ is the anchor selected within the k -step lookahead window and $\alpha \in (0, 1]$ balances coverage against movement consistency; the min keeps the movement-consistency term within $[0, 1]$ even when the anchor is not reachable within the window.

APPENDIX

A. Proof of Proposition 1

Proposition 1 (restated). *The exact trajectory recovery algorithm TREC returns the optimal trajectory P^* maximizing the overall coverage $\Omega(P)$ in $O(n|\mathcal{P}|^2)$ time, given the one-off pairwise shortest-path table on $\mathcal{M}$, where n is the number of timestamps and $|\mathcal{P}|$ is the number of candidate locations.*

Proof: For correctness, TREC ensures optimal trajectory reconstruction by applying the dynamic programming recurrence in Equation 1. At each timestamp t_i , the state value ω_{ij} for candidate p_j is updated from feasible predecessors, i.e., candidates p_k with $d_{\mathcal{M}}(p_j, p_k) \leq s \cdot (t_i - t_{i-1})$, thereby preserving Bellman’s principle of optimality [1]. The predecessor index recorded at each update allows complete reconstruction by backtracking from the endpoint maximizing ω_n , guaranteeing that the recovered trajectory P^* maximizes $\Omega(P)$ under all constraints.

For complexity, the nested loops over n timestamps and all $|\mathcal{P}|^2$ candidate pairs yield $O(n \cdot |\mathcal{P}|^2)$. Each distance check $d_{\mathcal{M}}(p_h, p_j)$ can be performed in $O(1)$ time after a standard preprocessing of candidate-pair distances, a routine step also adopted in related studies [2], [3]. Thus the overall runtime of TREC is $O(n \cdot |\mathcal{P}|^2)$. ■

B. Proof of Lemma 1

Lemma 1 (Spatial Constraint) (restated). *Every group θ of candidate points constructed by the grouping procedure of TRRP satisfies the diameter constraint $\max_{p_i, p_j \in \theta} d_{\mathcal{M}}(p_i, p_j) < \delta$, where δ is the given group diameter.*

Proof: We prove the spatial constraint property by analyzing the region formation process of the grouping procedure in TRRP. Consider any region θ generated by the procedure. The region starts with a single point p_i , where the maximum pairwise distance is trivially $0 \leq \delta$. Points are then added to θ iteratively. The condition for adding a point p_j to θ requires that for all existing points $p_k \in \theta$, the distance $d_{\mathcal{M}}(p_j, p_k) < \delta$. This ensures that after each addition, the maximum pairwise distance within θ remains strictly less than δ . Therefore, for the final region θ , $\max_{p_i, p_j \in \theta} d_{\mathcal{M}}(p_i, p_j) < \delta$. ■

C. Proof of Proposition 2

Proposition 2 (restated). *TRRP returns the recovered trajectory P' in $O(|\mathcal{P}|(\log |\mathcal{P}| + \nu) + nr^2)$ time, where n is the number of timestamps, $r = |\mathcal{R}|$ is the number of representative candidates, and ν bounds the number of candidates within any $\delta/2$ -neighborhood, determined by δ and the candidate granularity rather than by $|\mathcal{P}|$.*

Proof: TRRP first performs the grouping process. Each density score is obtained by a bounded-radius kd-tree range query of radius $\delta/2$, whose result is verified against the network distance, at a cost of $O(\log |\mathcal{P}| + \nu)$ per candidate; since each candidate is processed only once, the grouping complexity is $O(|\mathcal{P}|(\log |\mathcal{P}| + \nu))$. Each per-member distance test inside the group loop is answered by the same kd-tree structure at $O(\log |\mathcal{P}| + \nu_\delta)$ per query with $\nu_\delta = O(\nu)$ at fixed granularity, so grouping adds $O(|\mathcal{P}|(\log |\mathcal{P}| + \nu))$ overall. As an implementation convention, group-membership search is driven by kd-tree range queries of radius δ , at most a constant number per candidate, and at fixed granularity the number of candidates in a δ -neighborhood satisfies $\nu_\delta = O(\nu)$, so this search does not change the stated bound. Next, representatives are selected by computing the distance of each candidate to its group centroid, with total cost $O(|\mathcal{P}|)$. Finally, TREC is executed on the representative set with a complexity of $O(n \cdot r^2)$. Combining these stages, the overall complexity of the algorithm is $O(|\mathcal{P}|(\log |\mathcal{P}| + \nu) + n \cdot r^2)$. ■

D. Proof of Lemma 2

Lemma 2 (Bounded Coverage Variation) (restated). *Let $L := \max_{i \in [1, n]} \max_{\substack{p, q \in \mathcal{P} \\ p \neq q}} \frac{|\Omega_i(p) - \Omega_i(q)|}{d_{\mathcal{M}}(p, q)}$, which is well defined because $\mathcal{P}$ is finite and $d_{\mathcal{M}}(p, q) > 0$ for $p \neq q$. Then for any two candidates $p_j, p_k \in \mathcal{P}$, $|\Omega_i(p_j) - \Omega_i(p_k)| \leq L \cdot d_{\mathcal{M}}(p_j, p_k)$.*

Proof: Since $\mathcal{P}$ is finite and distinct candidates are distinct positions on $\mathcal{M}$ with $d_{\mathcal{M}}(p, q) > 0$, both maxima range over finite sets, so L is a well-defined finite constant. The bound then holds by construction: for any timestamp t_i and any $p_j \neq p_k$ in $\mathcal{P}$, the ratio $|\Omega_i(p_j) - \Omega_i(p_k)|/d_{\mathcal{M}}(p_j, p_k)$ appears in the maximum (in one of the two orders), hence is at most L ; for $p_j = p_k$ both sides vanish. Intuitively, L is controlled by the crossing density of coverage-region boundaries at the candidate granularity: the coverage count changes only where a path crosses such a boundary, so L

is governed by the density of base stations, the geometry of their coverage regions, and the sampling resolution of $\mathcal{P}$, and is independent of the timestamp and of the particular pair chosen. ■

E. Proof of Proposition 3

Proposition 3 (Coverage Bound) (restated). *For the recovered approximate trajectory P' , the coverage satisfies $\Omega(P') \geq \Omega(P^*) - (2n + \gamma n^2)L\delta$, where n is the number of timestamps, L is the constant of Lemma 2, δ is the group diameter, and γ is a constant arising from restoring speed-constraint feasibility after grouping.*

Proof: Let $P^* = \{p_1^*, \dots, p_n^*\}$ denote the optimal trajectory over $\mathcal{P}$, and for each timestamp i let $r_{\theta_i} \in \mathcal{R}$ denote the representative selected from the group θ_i containing p_i^* (here r_{θ_i} denotes a representative point, not the count $r = |\mathcal{R}|$). Since p_i^* and r_{θ_i} belong to the same group, Lemma 1 gives $d_{\mathcal{M}}(p_i^*, r_{\theta_i}) < \delta$, and Lemma 2 (both points are candidates) yields the per-timestamp replacement loss

$$\Omega_i(p_i^*) - \Omega_i(r_{\theta_i}) \leq L \cdot d_{\mathcal{M}}(p_i^*, r_{\theta_i}) \leq L\delta. \quad (3)$$

The projected sequence $\hat{P} = \{r_{\theta_1}, \dots, r_{\theta_n}\}$, however, is not necessarily feasible: by the triangle inequality, $d_{\mathcal{M}}(r_{\theta_i}, r_{\theta_{i+1}}) \leq d_{\mathcal{M}}(p_i^*, p_{i+1}^*) + 2\delta$, so consecutive representatives may exceed the speed budget by up to 2δ . We therefore construct a feasible surrogate trajectory $Q = \{q_1, \dots, q_n\} \subseteq \mathcal{R}$ inductively: set $q_1 = r_{\theta_1}$, and obtain q_{i+1} from q_i as follows. Follow a shortest path in $\mathcal{M}$ from q_i toward $r_{\theta_{i+1}}$, truncating it at arc length $s \cdot (t_{i+1} - t_i)$ if it is longer; retreat the resulting endpoint by $\delta + g$ along the path, where g is the candidate granularity, and snap the retreated point to its nearest representative w ; set $q_{i+1} = w$. Since candidates are sampled at granularity g , the retreated point lies within g of some candidate, whose group representative is within δ ; hence $d_{\mathcal{M}}(q_i, w) \leq s \cdot (t_{i+1} - t_i) - (\delta + g) + (\delta + g) = s \cdot (t_{i+1} - t_i)$, so reachability is preserved at every step. Because every candidate lies within distance δ of its representative, each step increases the residual deviation $d_{\mathcal{M}}(q_i, r_{\theta_i})$ by at most $c\delta$ for a constant c : at most 3δ is lost when approaching $r_{\theta_{i+1}}$ along a shortest path within the speed budget, at most 2δ when snapping to the nearest representative, and the retreat adds a further $\delta + g$, so relative to the naive construction c grows by $O(g/\delta)$; the form of the bound is unaffected and only γ is adjusted accordingly. Hence $d_{\mathcal{M}}(q_i, r_{\theta_i}) \leq c(i-1)\delta$. Combining both losses via Lemma 2 and the triangle inequality, and setting $\gamma := c/2$, we obtain

$$\Omega(P^*) - \Omega(Q) \leq \sum_{i=1}^n (\Omega_i(p_i^*) - \Omega_i(r_{\theta_i})) + (\Omega_i(r_{\theta_i}) - \Omega_i(q_i)) \quad (4)$$

$$\leq \sum_{i=1}^n (\delta + c(i-1)\delta) L \quad (5)$$

$$\leq (2n + \gamma n^2)L\delta. \quad (6)$$

When the remaining path length after q_i is shorter than $\delta + g$, the retreat is truncated at q_i itself and the snap target is q_i 's own group representative, so every proxy step remains feasible; the $O(g/\delta)$ term in c covers the residual snap drift in all cases.

Finally, Q is a feasible trajectory over the representative set $\mathcal{R}$, while TRRP returns the optimal trajectory P' over $\mathcal{R}$ by invoking TREC (Proposition 1). Therefore $\Omega(P') \geq \Omega(Q) \geq \Omega(P^*) - (2n + \gamma n^2)L\delta$. ■

F. Proof of Proposition 4

Proposition 4 (Approximation Guarantee) (restated). *TRRP returns an approximate trajectory P' such that $\Omega(P') \geq (1 - \epsilon)\Omega(P^*)$, where the group diameter is chosen as $\delta = \frac{\epsilon\Omega_{\max}}{(2n + \gamma n^2)L}$, in order to guarantee a $(1 - \epsilon)$ approximation.*

Proof: From Proposition 3, we have the coverage bound $\Omega(P') \geq \Omega(P^*) - (2n + \gamma n^2)L\delta$. Substituting the chosen grouping diameter $\delta = \frac{\epsilon\Omega_{\max}}{(2n + \gamma n^2)L}$ into the inequality yields $\Omega(P') \geq \Omega(P^*) - (2n + \gamma n^2)L \cdot \frac{\epsilon\Omega_{\max}}{(2n + \gamma n^2)L} = \Omega(P^*) - \epsilon\Omega_{\max}$. Since $\Omega(P^*) \geq \Omega_{\max}$, it follows that $\Omega(P') \geq \Omega(P^*) - \epsilon\Omega_{\max} \geq \Omega(P^*) - \epsilon\Omega(P^*) = (1 - \epsilon)\Omega(P^*)$. Thus, TRRP achieves a $(1 - \epsilon)$ approximation guarantee. ■

G. Proof of Proposition 5

Proposition 5 (restated). *TRHA returns the trajectory P' in $O(n|\mathcal{P}|)$ time after a one-off $O(|\mathcal{P}|^2)$ preprocessing of candidate-pair distances, where n is the number of timestamps and $|\mathcal{P}|$ is the number of candidate locations.*

Proof: First, the one-off preprocessing of candidate-pair distances requires $O(|\mathcal{P}|^2)$ time, after which each reachability check takes $O(1)$. With k -step lookahead, identifying the maximum-coverage candidates and selecting the anchor $\bar{p}_{i+k}$ both take $O(|\mathcal{P}|)$. For each candidate at t_i , computing the score in Equation 2 requires $O(|\mathcal{P}|)$ in total, and selecting the best candidate is subsumed in this bound. Thus each iteration costs $O(|\mathcal{P}|)$, leading to $O(n|\mathcal{P}|)$ across all timestamps. Together with the one-off preprocessing cost $O(|\mathcal{P}|^2)$, the total complexity is $O(|\mathcal{P}|^2 + n|\mathcal{P}|)$, i.e., $O(n|\mathcal{P}|)$ per trajectory after preprocessing. ■

H. Proof of Proposition 6

Proposition 6 (Coverage Bound of TRHA) (restated). *Given the trajectory $P' = \{p'_1, p'_2, \dots, p'_n\}$ obtained by TRHA with $\alpha \in (0, 1]$, and the optimal trajectory $P^* = \{p^*_1, p^*_2, \dots, p^*_n\}$, we have $\Omega(P') \geq \Omega(P^*) \cdot \min_{i \in [1, n]} \lambda_i - \frac{1 - \alpha}{\alpha} n C_i^{\max}$, where $\lambda_i = \frac{\max_{p \in \mathcal{F}_i(P'_{i-1})} \Omega_i(p)}{\max_{q \in \mathcal{F}_i(P^*_{i-1})} \Omega_i(q)}$ is the ratio of the best coverage attainable at t_i within the feasible sets $\mathcal{F}_i(\cdot)$ of the approximate and optimal solutions given their prefixes, and $C_i^{\max} := \max_{i \in [1, n]} C_i^{\max}$.*

Proof: Let $\mathcal{F}_i(P'_{i-1})$ be the set of feasible candidates at timestamp t_i given the prefix $\{p'_1, \dots, p'_{i-1}\}$ of the approximate solution, and similarly $\mathcal{F}_i(P^*_{i-1})$ for the optimal

solution. To measure the similarity between the feasible sets, we define the coverage ratio λ_i for each timestamp t_i as $\lambda_i = \frac{\max_{p \in \mathcal{F}_i(P'_{i-1})} \Omega_i(p)}{\max_{q \in \mathcal{F}_i(P^*_{i-1})} \Omega_i(q)}$.

Since $p^*_i \in \mathcal{F}_i(P^*_{i-1})$, we have $\max_{q \in \mathcal{F}_i(P^*_{i-1})} \Omega_i(q) \geq \Omega_i(p^*_i)$, and thus $\max_{p \in \mathcal{F}_i(P'_{i-1})} \Omega_i(p) \geq \lambda_i \cdot \Omega_i(p^*_i)$. If the denominator of λ_i is zero we set $\lambda_i = 1$; the corresponding timestamp then contributes equally (zero) to both feasible-set maxima and drops out of the bound. The algorithm selects a point $p'_i \in \mathcal{F}_i(P'_{i-1})$ by maximizing the score function in Equation 2, so $S_i(p'_i) \geq S_i(p)$ for any $p \in \mathcal{F}_i(P'_{i-1})$. In particular, let p be the point achieving the maximum coverage in $\mathcal{F}_i(P'_{i-1})$. By the min in Equation 2, the movement-consistency term $(1 - \alpha) \cdot (1 - \min\{1, d_{\mathcal{M}}(p, \bar{p}_{i+k})/(s \cdot (t_{i+k} - t_i))\})$ lies in $[0, 1 - \alpha]$ for every candidate, regardless of whether the anchor $\bar{p}_{i+k}$ is reachable within the lookahead window; it may therefore be dropped from the right-hand side unconditionally, yielding:

$$\alpha \cdot \frac{\Omega_i(p'_i)}{C_i^{\max}} + (1 - \alpha) \cdot \left(1 - \min\left\{1, \frac{d_{\mathcal{M}}(p'_i, \bar{p}_{i+k})}{s \cdot (t_{i+k} - t_i)}\right\}\right) \geq \alpha \cdot \frac{\Omega_i(p)}{C_i^{\max}}. \quad (7)$$

Rearranging and combining with the earlier inequality for $\max_{p \in \mathcal{F}_i(P'_{i-1})} \Omega_i(p)$, we get

$$\Omega_i(p'_i) \geq \lambda_i \cdot \Omega_i(p^*_i) - \frac{(1 - \alpha)}{\alpha} C_i^{\max} \cdot \left(1 - \min\left\{1, \frac{d_{\mathcal{M}}(p'_i, \bar{p}_{i+k})}{s \cdot (t_{i+k} - t_i)}\right\}\right). \quad (8)$$

Each parenthesized term lies in $[0, 1]$ by construction. For the last k timestamps no anchor is applied, so the score reduces to the coverage term and $\Omega_i(p'_i) \geq \lambda_i \cdot \Omega_i(p^*_i)$ holds directly without penalty. Summing over $i = 1$ to n therefore yields

$$\Omega(P') = \sum_{i=1}^n \Omega_i(p'_i) \quad (9)$$

$$\geq \sum_{i=1}^n \lambda_i \cdot \Omega_i(p^*_i) - \frac{(1 - \alpha)}{\alpha} \sum_{i=1}^{n-k} C_i^{\max} \cdot \left(1 - \min\left\{1, \frac{d_{\mathcal{M}}(p'_i, \bar{p}_{i+k})}{s \cdot (t_{i+k} - t_i)}\right\}\right) \quad (10)$$

$$\geq \Omega(P^*) \cdot \min_{i \in [1, n]} \lambda_i - \frac{(1 - \alpha)}{\alpha} n C_i^{\max}, \quad (11)$$

where $C_i^{\max} = \max_{i \in [1, n]} C_i^{\max}$. ■

I. Proof of Proposition 7

Proposition 7 (restated). *Replacing $\mathcal{M}$ with the localized subnetwork $\mathcal{M}'$ preserves the optimal trajectory. Specifically, solving the recovery problem on $\mathcal{M}'$ yields a trajectory P' satisfying the speed constraint with $\Omega(P') = \Omega(P^*)$, where P^* is the optimal solution on $\mathcal{M}$.*

Proof: Coverage is preserved since each $p_i \in D_c$ for some $c \in C_i$ belongs to $U \subseteq \mathcal{M}'$, so $\Omega_i(p_i)$ takes the same value in $\mathcal{M}$ and $\mathcal{M}'$. Feasibility is preserved because for any consecutive points p_i, p_{i+1} , a shortest path π in $\mathcal{M}$ can be decomposed into segments either inside R or

outside with endpoints on ∂R . Internal segments are included in $\mathcal{M}'$, and for external segments, shortest paths between boundary points are preserved. Replacing each segment by its preserved path yields a path in $\mathcal{M}'$ of the same length, so $d_{\mathcal{M}'}(p_i, p_{i+1}) = d_{\mathcal{M}}(p_i, p_{i+1})$ and the speed constraint is maintained.

A subtlety concerns zero-coverage “stepping-stone” points: a feasible hop may traverse positions covered by no observed station, and the argument above must preserve not only the hop’s endpoints but also the existence of an equal-length route in $\mathcal{M}'$. The segment decomposition provides it directly: every $\mathcal{M}$ -shortest path between two candidates is re-expressed in $\mathcal{M}'$ at the same length, so intermediate positions, covered or not, never need to be represented explicitly by candidates; where the surrogate construction of Proposition 3 places a candidate at arc length $s \cdot \Delta t_1$ along the route, the same replacement applies on $\mathcal{M}'$, and the boundary-distance equalities hold up to the candidate discretization slack, which is absorbed by the δ -terms of Propositions 3 and 4. At the sampled granularity a replacement candidate exists by construction of $\mathcal{P}$; if the exact arc-length point falls between samples, the nearest candidate adds $O(g)$ drift, absorbed into the Lemma 2 constant.

We do not claim that the recovered point sequence is literally the same object computed on $\mathcal{M}$; rather, the DP over the candidate set with distances $d_{\mathcal{M}'}$ admits exactly the same feasible transitions and the same coverage values as the DP with $d_{\mathcal{M}}$, so an optimal solution on $\mathcal{M}'$ is an optimal solution on $\mathcal{M}$ (under the tie-breaking rule of the main paper, the same one is selected). For the approximation guarantees, note that $\Omega(P^*)$, the loss term $(2n + \gamma n^2)L\delta$, the ratios λ_i , and $C^{\max}$ are all maxima over the candidate set, and the candidate set restricted to $\mathcal{M}'$ is $\mathcal{P} \cap \mathcal{M}' = \{p \in \mathcal{P} \mid p \in U \text{ or } p \text{ lies on a preserved boundary path}\}$, which contains every candidate that any feasible trajectory can visit with positive coverage; candidates outside contribute zero coverage and are dominated. The bounds of Propositions 3 and 6 therefore hold verbatim on $\mathcal{M}'$: for TRRP the group diameter δ , the constant L of Lemma 2 (a maximum over candidate pairs), and $\Omega_{\max}$ are unchanged when the maximum is restricted to $\mathcal{P} \cap \mathcal{M}'$; for TRHA the anchors $\bar{p}_{i+k}$ are maximum-coverage candidates and hence lie in $\mathcal{P} \cap \mathcal{M}'$, and the min-form of the score makes the movement term well defined and in $[0, 1]$ for every candidate, so the feasible-set ratios λ_i are unchanged. ■

J. Proof of Proposition 8

Proposition 8 (restated). *After applying TRSN optimization, the time complexity becomes $O(\sum_{\theta=1}^{|\Theta|} M_{\theta} n \eta_{\theta}^2 |\mathcal{P}|^2)$ for TREC; $O(\sum_{\theta=1}^{|\Theta|} M_{\theta} (\eta_{\theta} |\mathcal{P}| (\log |\mathcal{P}| + \nu) + n_{\theta} \eta_{\theta}^2 r_{\theta}^2))$ for TRRP, with r_{θ} the realized representative count; and $O(\sum_{\theta=1}^{|\Theta|} M_{\theta} n \eta_{\theta} |\mathcal{P}|)$ for TRHA, where $|\Theta|$ is the number of subnetworks, M_{θ} is the number of sequences in subnetwork θ , n is the maximum number of timestamps among the sequences in θ , n_{θ} is the maximum number of timestamps among the sequences*

assigned to subnetwork θ ($n_{\theta} \leq n$), $|\mathcal{P}|$ is the original candidate set size, r is the number of representatives, and η_{θ} (with $0 < \eta_{\theta} < 1$) is the reduction ratio per subnetwork.

Proof: The time complexity analysis follows from applying the TRSN optimization to each algorithm’s base complexity. The key insight is that by partitioning the problem into independent subnetworks, we can analyze the complexity per subnetwork and then sum over all subnetworks.

For TREC, the base complexity is $O(n|\mathcal{P}|^2)$. After partitioning, for each subnetwork θ with M_{θ} cellular record sequences and candidate set size reduced to $\eta_{\theta}|\mathcal{P}|$ due to spatial localization, the complexity becomes $O(M_{\theta} n \eta_{\theta}^2 |\mathcal{P}|^2)$. Summing over all subnetworks gives $O(\sum_{\theta=1}^{|\Theta|} M_{\theta} n \eta_{\theta}^2 |\mathcal{P}|^2)$. For TRRP, after partitioning, the grouping complexity per subnetwork is $O(\eta_{\theta} |\mathcal{P}| (\log |\mathcal{P}| + \nu))$ (kd-tree membership queries per candidate) and the recovery over the shrunken representative set is $O(n_{\theta} \eta_{\theta}^2 r_{\theta}^2)$, where r_{θ} denotes the realized number of representatives in subnetwork θ (empirically $r_{\theta} \leq \eta_{\theta} r$). Applied to M_{θ} sequences, the total complexity becomes $O(\sum_{\theta=1}^{|\Theta|} M_{\theta} (\eta_{\theta} |\mathcal{P}| (\log |\mathcal{P}| + \nu) + n_{\theta} \eta_{\theta}^2 r_{\theta}^2))$. The per-subnetwork distance tables are precomputed offline once and amortized over all trajectories, following the same accounting convention as the stage-timing table (Table I), and are not counted per trajectory. For TRHA, the base algorithm has complexity $O(n|\mathcal{P}|)$. After partitioning, the per subnetwork complexity is $O(n \eta_{\theta} |\mathcal{P}|)$ for M_{θ} sequences, giving total complexity $O(\sum_{\theta=1}^{|\Theta|} M_{\theta} n \eta_{\theta} |\mathcal{P}|)$.

The reduction ratios η_{θ} , $0 < \eta_{\theta} < 1$ capture the spatial localization effect, representing the fraction of the original candidate set that remains relevant within each subnetwork θ . ■

Figures 4 and 5 report sensitivity to the candidate granularity and to the horizon window size k ; **TREC and TRHA stay within 0.03 km and 0.18 km respectively across granularities 0.1–1.0 km, whereas TRRP degrades beyond 0.4 km as groups merge distinct road segments; the window size minimizes RMSE at $k = 2$, where a short look-ahead already balances instantaneous coverage against increasingly uncertain long-term information.**

K. Stage-by-stage Timing

Table I breaks the per-trajectory recovery time into candidate-structure construction (Build) and the main loop (Recover), complementing the end-to-end times in the main paper.

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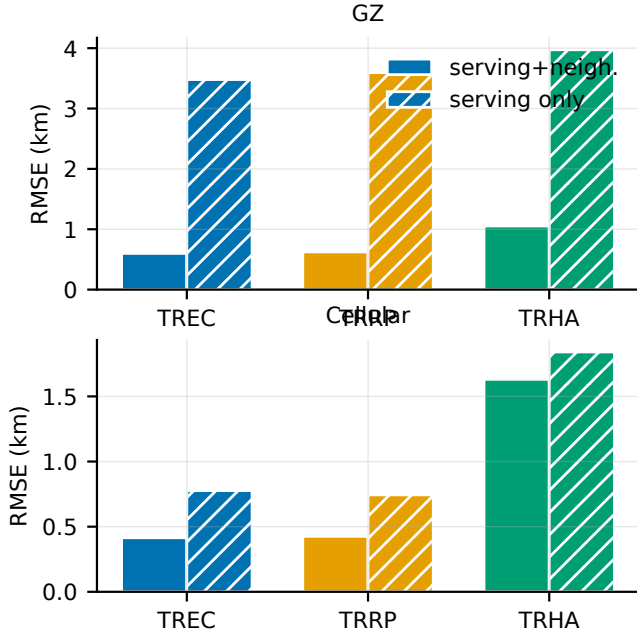


Fig. 1. Serving-only ablation: restricting each record to its serving station versus using serving and neighboring stations jointly.

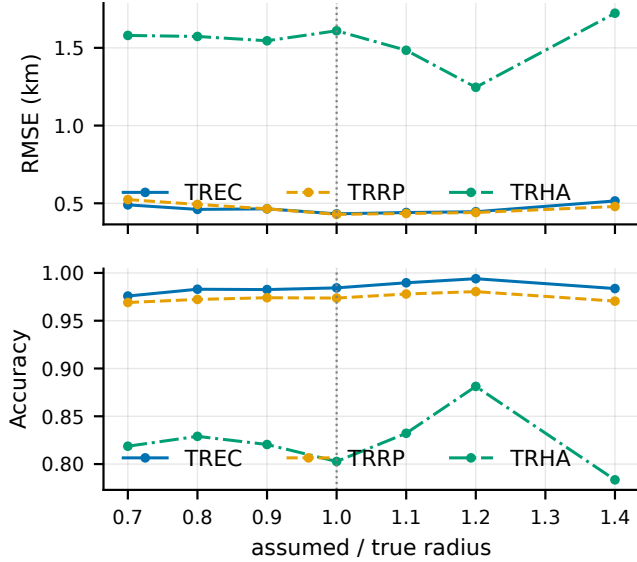
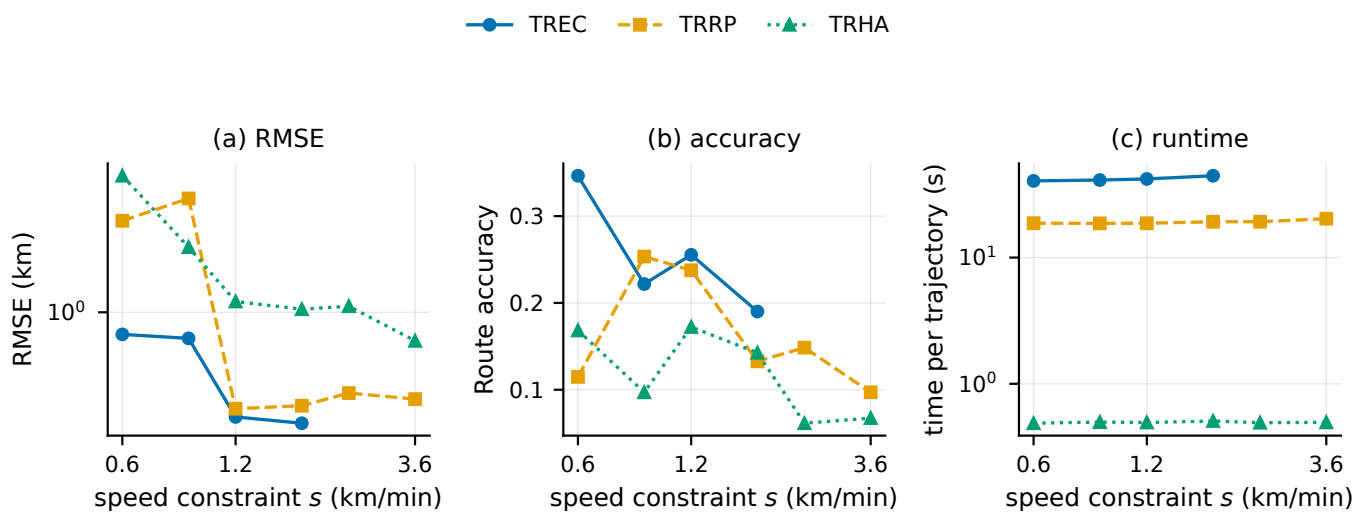


Fig. 2. Sensitivity to coverage-region mismatch on Cellular: assumed radius scaled relative to the true radius (vertical line).

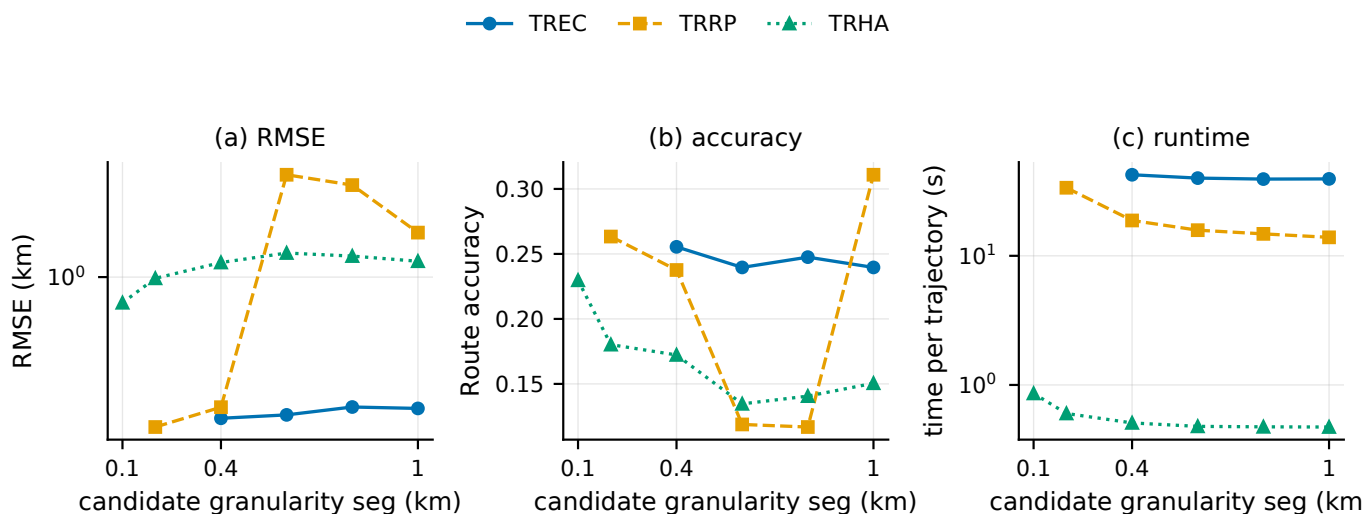
TABLE I
STAGE-BY-STAGE TIMING OF THE THREE RECOVERY ALGORITHMS ON GZ-4 AND GZ-5 (MEAN PER QUERY TRAJECTORY, FULL-OBSERVATION SETTING). BUILD = CANDIDATE-STRUCTURE CONSTRUCTION (COVERAGE SCORING, FEASIBLE-TRANSITION GRAPH FOR TREC; CLUSTERING, GRAPH COMPRESSION AND COMPRESSED DISTANCE MATRIX FOR TRRP; OUT-EDGE AND COVERAGE PRECOMPUTATION FOR TRHA); RECOVER = THE DP MAIN LOOP (TREC, AND THE INNER DP OF TRRP) OR THE GREEDY MAIN LOOP (TRHA). PREPROCESSING (ROAD-NETWORK LOADING AND FULL-NETWORK DISTANCE MATRIX, 37.5 s FOR GZ-4 AND 24.4 s FOR GZ-5) IS A *one-time offline* COST SHARED BY ALL ALGORITHMS AND ALL TRAJECTORIES; FOLLOWING COMMON PRACTICE IN THE FIELD, IT IS AMORTIZED OVER THE OFFLINE PHASE AND *not* COUNTED IN THE PER-TRAJECTORY TIMES (BUILD/RECOVER/TOTAL). STAGE TIMINGS WERE COLLECTED WITH INSTRUMENTED COPIES OF THE RELEASED IMPLEMENTATIONS, VERIFIED TO RETURN IDENTICAL RECOVERY RESULTS; THE PER-DATASET SUMS OF BUILD AND RECOVER MATCH THE PUBLISHED END-TO-END PER-TRAJECTORY TIMES WITHIN 1.9% FOR TREC AND TRHA, WHILE TRRP'S INSTRUMENTED BUILD RECOMPRESSES PER SEGMENT WITHOUT THE RELEASED PIPELINE'S CROSS-SEGMENT CACHE REUSE.

Dataset	Algorithm	Build (ms)	Recover (ms)	Total (ms)
<i>GZ-4 (30 trajectories), one-time preprocessing 37.5 s</i>				
	TREC	45,498.0	1,978.2	47,476.2
	TRRP	20,561.3	150.5	20,711.9
	TRHA	501.1	24.9	526.0
<i>GZ-5 (10 trajectories), one-time preprocessing 24.4 s</i>				
	TREC	24,581.8	1,221.4	25,803.1
	TRRP	12,527.0	81.4	12,608.4
	TRHA	390.2	17.8	408.0



dropped (20-min cap): $s=2.4$ TREC; $s=3.6$ TREC | kept despite >20 min ($\geq 80\%$ segs done): TREC

Fig. 3. Sensitivity to the speed constraint s over GZ; the vertical line marks the operating point $s = 1.2$ km/min. TREC completes all 40 segments for $s \leq 1.8$ but 10/40 at $s = 2.4$ and 3.6; partial runs are excluded.



dropped (20-min cap): seg=0.1 TREC; seg=0.1 TRRP; seg=0.2 TREC | kept despite >20 min ($\geq 80\%$ segs done): TREC

Fig. 4. Trajectory recovery performance for varying the candidate granularity over GZ; larger values correspond to sparser spatial sampling along the road network.

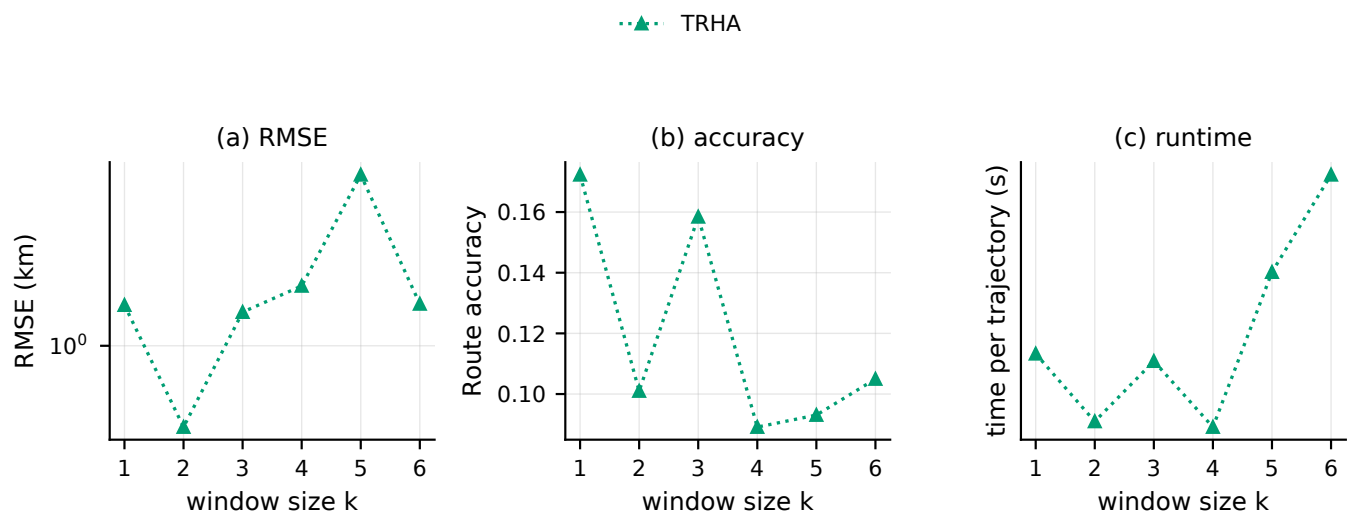


Fig. 5. Trajectory recovery performance for varying the window size k over GZ; **RMSE is minimized at $k = 2$, while route accuracy is highest at $k = 1$ and spans 8.3 percentage points across the swept window sizes.**